Mathematical Equations of Neural Circuits and Networks

Chapter 5: Computational Models of Neural Circuit Organization

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Abstract

This document presents the mathematical foundation for understanding neural circuits and networks. Every major circuit principle described in Chapter 5 is converted into published computational models — covering lateral inhibition, attractor networks, E-I balance, large-scale connectivity, cortico-basal ganglia loops, cerebellar computation, reward circuits, fear conditioning, and hippocampal memory. These models span from local microcircuits to whole-brain network dynamics.

Contents

1 Circuit Organization Principles

1.1 Lateral Inhibition — Mexican-Hat Kernel

What it models: Surround inhibition in sensory cortex. A neuron excited by a stimulus simultaneously inhibits its neighbors, sharpening the representation. The net connectivity profile is excitatory near the center and inhibitory in the surround — the "Mexican hat" shape.

Ref: Rodieck, R.W. (1965). "Quantitative analysis of cat retinal ganglion cell response to visual stimuli." *Vision Research*, 5(11), 583–601.

1.1.1 Difference-of-Gaussians (DoG) Receptive Field

$$w(x) = A_e \exp\left(-\frac{x^2}{2\sigma_e^2}\right) - A_i \exp\left(-\frac{x^2}{2\sigma_i^2}\right) \tag{1}$$

1.1.2 Network Response with Lateral Inhibition

$$r_i(t+1) = f\left(\sum_j w(x_i - x_j)\, r_j(t) + I_i\right) \tag{2}$$

Variable	Definition	Typical Values
$w(x)$	Synaptic weight as function of distance	—
A_e, A_i	Excitatory/inhibitory amplitudes	$A_e > A_i$
σ_e, σ_i	Excitatory/inhibitory widths	$\sigma_e < \sigma_i$
$x_i - x_j$	Distance between neurons i and j	mm or degrees
$f(\cdot)$	Activation function (sigmoid or ReLU)	—
I_i	External input to neuron i	—

1.1.3 Sharpening Effect

$$\mathrm{SNR}_{\mathrm{output}} = \frac{r_{\mathrm{center}}}{r_{\mathrm{surround}}} \propto \frac{A_e/\sigma_e^2}{A_i/\sigma_i^2} \tag{3}$$

Key insight: Lateral inhibition is the circuit mechanism behind contrast enhancement in vision (Mach bands), two-point discrimination in touch, and frequency sharpening in the cochlea. The DoG profile emerges from the balance between fast short-range excitation and slower long-range inhibition.

1.2 Recurrent Attractor Network — Hopfield Energy Function

What it models: Recurrent networks (like CA3 hippocampus or PFC working memory) have stable attractor states corresponding to stored memories or maintained representations. The network "settles" into the nearest attractor.

Ref: Hopfield, J.J. (1982). "Neural networks and physical systems with emergent collective computational abilities." *PNAS*, 79(8), 2554–2558.

1.2.1 State Update Rule

$$s_i(t+1) = \text{sign}\left(\sum_j w_{ij} s_j(t)\right) \tag{4}$$

1.2.2 Synaptic Weight Matrix (Hebbian Learning)

$$w_{ij} = \frac{1}{N}\sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu, \quad w_{ii} = 0 \tag{5}$$

1.2.3 Energy Function

$$E = -\frac{1}{2}\sum_{i \neq j} w_{ij} s_i s_j \tag{6}$$

Variable	Definition
$s_i \in \{-1, +1\}$	State of neuron i (inactive/active)
w_{ij}	Symmetric weight (excitatory + excitatory = positive)
$\xi_i^\mu \in \{-1, +1\}$	Value of neuron i in stored pattern μ
p	Number of stored patterns
N	Total number of neurons
E	Network energy (always decreases during retrieval)

1.2.4 Storage Capacity

$$p_{\max} \approx 0.138\, N \tag{7}$$

Key insight: The network can store approximately 14% of N patterns. Recall is perfect when starting from a partial cue (at least ˜50% correct). Above capacity, retrieval fails with "spurious attractors." This capacity limit maps onto biological memory interference and forgetting.

1.3 Wilson-Cowan E-I Balance and Oscillations

What it models: Cortical columns containing excitatory (E) and inhibitory (I) populations. The balance between E and I determines whether the network is at a stable point, oscillating (gamma), or bistable (persistent activity).

Ref: Wilson, H.R. & Cowan, J.D. (1972). *Biophysical Journal*, 12(1), 1–24.

1.3.1 Coupled Population Equations

$$\tau_E \frac{d\bar{E}}{dt} = -\bar{E} + (1 - r_E \bar{E})\, S_E\big[c_1 \bar{E} - c_2 \bar{I} + P\big] \tag{8}$$

$$\tau_I \frac{d\bar{I}}{dt} = -\bar{I} + (1 - r_I \bar{I})\, S_I\big[c_3 \bar{E} - c_4 \bar{I} + Q\big] \tag{9}$$

1.3.2 Sigmoid Activation

$$S(x) = \frac{1}{1 + e^{-a(x-\theta)}} \tag{10}$$

1.3.3 Stability Condition (E-I Balance)

For a stable fixed point, the Jacobian eigenvalues must have negative real parts:

$$\lambda_{1,2} = \frac{1}{2}\left[(J_{EE} + J_{II}) \pm \sqrt{(J_{EE} - J_{II})^2 + 4J_{EI}J_{IE}}\right] \tag{11}$$

where $J_{EE} = c_1 S'_E$, $J_{EI} = -c_2 S'_I$, $J_{IE} = c_3 S'_E$, $J_{II} = -c_4 S'_I$.
Oscillatory condition (gamma rhythm): When $\mathrm{Re}(\lambda) = 0$, imaginary λ produces oscillations at frequency:

$$f_\gamma = \frac{1}{2\pi}\mathrm{Im}(\lambda) \approx 30\text{–}80 \text{ Hz (gamma band)} \tag{12}$$

Parameter	Definition	Role
c_1	E→E recurrence	Working memory
c_2	I→E feedback inhibition	Gamma oscillation, stability
c_3	E→I feedforward	Gain control
c_4	I→I	Prevents runaway inhibition

1.4 VIP→SST→Pyramidal Disinhibitory Circuit

What it models: The 3-population disinhibitory motif: VIP+ interneurons inhibit SST+ interneurons, releasing pyramidal cells from SST-mediated dendritic inhibition. This circuit implements top-down attentional gain modulation.

Ref: Pfeffer, C.K., Xue, M., He, M., et al. (2013). "Inhibition of inhibition in visual cortex: the logic of connections between molecularly distinct interneurons." *Nature Neuroscience*, 16(8), 1068–1076.

1.4.1 3-Population Rate Model

$$\tau_P \frac{dr_P}{dt} = -r_P + f(w_{EP}r_E - w_{SP}r_S - w_{PP}r_P + I_P) \tag{13}$$

$$\tau_S \frac{dr_S}{dt} = -r_S + f(w_{ES}r_E - w_{VS}r_V) \tag{14}$$

$$\tau_V \frac{dr_V}{dt} = -r_V + f\left(w_{EV}r_E + I_V^{\text{top-down}}\right) \tag{15}$$

Variable	Definition
r_P, r_S, r_V	Firing rates: Pyramidal, SST+, VIP+
w_{SP}	SST→Pyramid (dendritic inhibition)
w_{VS}	VIP→SST (disinhibition pathway)
w_{EP}, w_{ES}, w_{EV}	Excitatory drive to each population
$I_V^{\text{top-down}}$	Top-down input activating VIP neurons

1.4.2 Net Effect of Top-Down Input

$$\Delta r_P = +\frac{w_{SP} \cdot w_{VS}}{w_{VS}^2} \cdot I_V^{\text{top-down}} \quad \text{(disinhibitory gain)} \tag{16}$$

Key insight: When top-down attention activates VIP neurons → VIP inhibits SST → SST inhibition of pyramidal dendrites is removed → pyramidal cell gain increases. This is the circuit mechanism for attentional amplification of task-relevant cortical representations.

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2 Large-Scale Network Metrics

2.1 Functional Connectivity — Pearson Correlation

What it models: How strongly the BOLD fMRI time series of two brain regions covary. Positive correlation = functional connection within a network (e.g., DMN). Negative correlation = anti-correlation between competing networks (DMN vs CEN).

Ref: Biswal, B., Yetkin, F.Z., Haughton, V.M. & Hyde, J.S. (1995). "Functional connectivity in the motor cortex of resting human brain using echo-planar MRI." *Magnetic Resonance in Medicine*, 34(4), 537–541.

2.1.1 Pearson Correlation Coefficient

$$r_{ij} = \frac{\sum_{t=1}^{T}(x_i(t) - \bar{x}_i)(x_j(t) - \bar{x}_j)}{\sqrt{\sum_t (x_i(t) - \bar{x}_i)^2 \sum_t (x_j(t) - \bar{x}_j)^2}} \tag{17}$$

2.1.2 Fisher Z-Transform (for Statistical Testing)

$$Z_{ij} = \frac{1}{2}\ln\left(\frac{1 + r_{ij}}{1 - r_{ij}}\right) = \operatorname{arctanh}(r_{ij}) \tag{18}$$

Variable	Definition
$x_i(t)$	BOLD signal time series at region i
T	Number of time points (~300–600 for 10-min scan)
r_{ij}	Functional connectivity between i and j (-1 to $+1$)
Z_{ij}	Fisher-transformed FC (approximately normal)

Typical values: Same-network pairs: $r \approx 0.3$–0.7; Anti-correlated networks (DMN↔CEN): $r \approx -0.2$–-0.4

2.2 Graph Theory — Small-World Network Metrics

What it models: The brain's large-scale network topology has "small-world" properties: high local clustering (specialized modules) AND short path lengths (efficient global integration). This is quantified by Watts-Strogatz metrics.

Ref: Watts, D.J. & Strogatz, S.H. (1998). "Collective dynamics of 'small-world' networks." *Nature*, 393(6684), 440–442.

2.2.1 Clustering Coefficient

$$C_i = \frac{2e_i}{k_i(k_i - 1)} \tag{19}$$

$$C = \frac{1}{N}\sum_{i=1}^{N} C_i \tag{20}$$

2.2.2 Characteristic Path Length

$$L = \frac{1}{N(N-1)}\sum_{i \neq j} d_{ij} \tag{21}$$

2.2.3 Small-World Index

$$\sigma = \frac{C/C_{\text{random}}}{L/L_{\text{random}}} \tag{22}$$

Small-world criterion: $\sigma > 1$ (higher clustering than random, similar path length)

Variable	Definition
e_i	Number of edges between neighbors of node i
k_i	Degree of node i (number of connections)
d_{ij}	Shortest path length between nodes i and j
C_{random}	Clustering of equivalent random network
L_{random}	Path length of equivalent random network

2.2.4 Degree Distribution (Scale-Free Hubs)

$$P(k) \propto k^{-\gamma}, \quad \gamma \approx 2\text{–}3 \text{ (brain hubs)} \tag{23}$$

Hub nodes (high degree) in the brain: PCC/precuneus, mPFC, insula, thalamus — overlap with DMN and salience network core nodes.
Key insight: The brain's small-world topology minimizes wiring cost (short paths) while maximizing local processing efficiency (high clustering). Hub nodes like PCC and mPFC are highly connected to multiple networks — explaining why DMN disruption in Alzheimer's spreads to multiple systems.

2.3 Dynamic Functional Connectivity — Sliding Window

What it models: FC is not static — it fluctuates over seconds-to-minutes, reflecting switching between cognitive states. Sliding window analysis captures these dynamics.

Ref: Chang, C. & Glover, G.H. (2010). "Time-frequency dynamics of resting-state brain connectivity measured with fMRI." *NeuroImage*, 50(1), 81–98.

2.3.1 Windowed Correlation

$$r_{ij}^{(t_0)} = r(x_i[t_0 : t_0 + W],\ x_j[t_0 : t_0 + W]) \tag{24}$$

2.3.2 State Identification (k-means Clustering)

Concatenate all windowed FC matrices and cluster into k states:

$$\hat{S}(t) = \arg\min_k \|\mathbf{FC}(t) - \boldsymbol{\mu}_k\|^2 \tag{25}$$

Parameter	Definition	Typical Values
W	Window width	30–60 s (15–30 TRs at TR=2s)
k	Number of FC states	3–6 (data-driven)
$\boldsymbol{\mu}_k$	Centroid FC matrix of state k	—

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3 Sensory Processing Circuits

3.1 Visual Pathway — Magnocellular vs Parvocellular Streams

What it models: LGN relay separates visual information into M (motion/contrast) and P (color/detail) streams. Each has distinct temporal and spatial filtering properties.

Ref: Enroth-Cugell, C. & Robson, J.G. (1966). "The contrast sensitivity of retinal ganglion cells of the cat." *Journal of Physiology*, 187(3), 517–552.

3.1.1 Center-Surround Receptive Field (DoG)

$$RF(x, y) = \frac{1}{2\pi\sigma_c^2} e^{-(x^2+y^2)/2\sigma_c^2} - \frac{k}{2\pi\sigma_s^2} e^{-(x^2+y^2)/2\sigma_s^2} \tag{26}$$

Parameter	M pathway	P pathway
σ_c (center)	0.5–2	0.05–0.5
σ_s (surround)	2–8	0.2–2
Temporal resolution	High (>50 Hz)	Low (<20 Hz)
Contrast sensitivity	High	Moderate

3.1.2 Contrast Sensitivity Function (CSF)

$$\mathrm{CSF}(f) = a \cdot f^b \cdot e^{-cf} \tag{27}$$

where f is spatial frequency (cycles/degree). Peak sensitivity at $\sim$ 3–5 cpd for humans.

3.2 Thalamocortical Relay — Effective Connectivity

What it models: The thalamus gates sensory information based on behavioral state. During attention, corticothalamic feedback (Layer VI→thalamus) increases thalamic relay gain; during inattention/sleep, reticular nucleus inhibition reduces relay fidelity.

$$r_{\mathrm{TC}}(t) = G_{\mathrm{relay}} \cdot r_{\mathrm{input}}(t) - G_{\mathrm{RE}} \cdot r_{\mathrm{RE}}(t) \tag{28}$$

$$G_{\mathrm{relay}} = G_0 + \Delta G \cdot r_{\mathrm{ctx}}^{\mathrm{feedback}}(t) \tag{29}$$

Variable	Definition
r_{TC}	Thalamocortical relay neuron firing rate
G_{relay}	Relay gain (modulated by cortical feedback)
r_{RE}	Reticular nucleus inhibitory rate
ΔG	Attentional gain modulation ($\sim$ 20–40%)

4 Cortico-Basal Ganglia-Thalamo-Cortical Loops

4.1 Direct and Indirect Pathway — Rate Model

What it models: The two competing BG pathways: Direct (D1, "Go" — facilitates action) and Indirect (D2, "NoGo" — suppresses action). The net output (GPi/SNr) is the algebraic difference, gating thalamic activity.

Ref: Albin, R.L., Young, A.B. & Penney, J.B. (1989). "The functional anatomy of basal ganglia disorders." *Trends in Neurosciences*, 12(10), 366–375.

4.1.1 GPi/SNr Output (Action Selection)

$$r_{\text{GPi}} = r_{\text{GPi,base}} - w_D r_{\text{D1,striatum}} + w_I r_{\text{D2,striatum}} + w_{\text{STN}} r_{\text{STN}} \tag{30}$$

4.1.2 Thalamic Disinhibition

$$r_{\text{thal}} = r_{\text{thal,base}} - w_{\text{GPi}} r_{\text{GPi}} \tag{31}$$

$$r_{\text{cortex}} = f(r_{\text{thal}}) \tag{32}$$

4.1.3 Dopamine Modulation

$$r_{\text{D1,striatum}} = r_{\text{str}} \cdot (1 + k_{D1}[\text{DA}]) \tag{33}$$

$$r_{\text{D2,striatum}} = r_{\text{str}} \cdot (1 - k_{D2}[\text{DA}]) \tag{34}$$

Parameter	Definition	Typical Values
$r_{\text{GPi,base}}$	Baseline GPi firing (high at rest)	60–80 Hz
w_D, w_I	Direct/indirect pathway weights	—
k_{D1}, k_{D2}	DA modulation gains on D1/D2	0.1–0.5
r_{STN}	Subthalamic nucleus (hyperdirect)	—

Parkinson's model: DA loss $\rightarrow$ reduced D1 (less Go) and less D2 suppression (more NoGo) $\rightarrow$ increased GPi firing $\rightarrow$ more thalamic suppression $\rightarrow$ hypokinesia.

4.2 TD Actor-Critic — Full BG Implementation

What it models: Basal ganglia implements reinforcement learning. Striatum = Actor (selects actions) + Critic (evaluates states). Dopamine = TD error signal.

Ref: Barto, A.G. (1995). In Houk, Davis & Beiser (Eds.), *Models of Information Processing in the Basal Ganglia*, MIT Press.

4.2.1 TD Error (Dopamine Signal)

$$\delta(t) = r(t) + \gamma V(s_{t+1}) - V(s_t) \tag{35}$$

4.2.2 Critic Update (Striatal Synapses, D1+D2)

$$V(s_t) \leftarrow V(s_t) + \alpha_c \delta(t) \tag{36}$$

4.2.3 Actor Update — Direct Path (D1, Go)

$$\Delta w_{\text{Go}}(s, a) = \alpha_a [\delta]^+ \cdot e(s, a) \tag{37}$$

4.2.4 Actor Update — Indirect Path (D2, NoGo)

$$\Delta w_{\text{NoGo}}(s, a) = \alpha_a [-\delta]^+ \cdot e(s, a) \tag{38}$$

4.2.5 STN Hyperdirect — Global NoGo

$$\text{STN}_{\text{out}} = \beta \sum_a P(\text{conflict}_a) \quad \text{(raises decision threshold globally)} \tag{39}$$

Variable	Definition
$\delta(t)$	TD error $\rightarrow$ DA burst ($\delta > 0$) or dip ($\delta < 0$)
$[\cdot]^+$	Positive part (rectification)
$e(s, a)$	Eligibility trace for state-action pair
α_c, α_a	Learning rates for critic and actor
γ	Temporal discount factor

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5 Cerebellar Circuit Computation

5.1 Marr-Albus-Ito Adaptive Filter

What it models: The cerebellum as a supervised learning system. Parallel fibers carry contextual signals (mossy fiber input); climbing fibers carry teaching signal (motor error). Purkinje cell output = motor correction.

Refs: Marr 1969; Albus 1971; Fujita, M. (1982). *Biological Cybernetics*, 45(3), 195–206.

5.1.1 Purkinje Cell Output

$$y(t) = \sum_{i=1}^{N} w_i(t)\, x_i(t) \tag{40}$$

5.1.2 Error-Driven LTD at Parallel Fiber Synapses

$$\frac{dw_i}{dt} = -\eta\, x_i(t)\, e(t) \tag{41}$$

where $e(t)$ = climbing fiber error signal from inferior olive.

5.1.3 Motor Error Convergence

$$e(t) = d(t) - y(t) \tag{42}$$

$$\frac{d}{dt}\|e\|^2 = -2\eta \sum_i x_i^2 \|e\|^2 < 0 \tag{43}$$

(Error always decreases → learning is stable)

Variable	Definition
$y(t)$	Purkinje cell output (motor correction)
$w_i(t)$	Parallel fiber → Purkinje cell synaptic weight
$x_i(t)$	Granule cell (parallel fiber) activity
N	Number of parallel fibers ($\sim$ 200,000)
$e(t)$	Motor error (climbing fiber signal)
$d(t)$	Desired output (correct movement)
η	Learning rate

5.1.4 Timing — ISI Refinement

$$\hat{t}_{\text{peak}} = \frac{\sum_i x_i(t) w_i t_i}{\sum_i x_i(t) w_i} \tag{44}$$

Purkinje cell adjusts timing of conditioned eye-blink within millisecond precision by weighting granule cell inputs by their onset times.

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6 Reward Circuitry

6.1 NAc Core/Shell — Incentive Salience (Wanting vs Liking)

What it models: Berridge's distinction between "wanting" (incentive salience, NAc core + DA) and "liking" (hedonic pleasure, NAc shell opioid hotspots). These dissociate in addiction: high "wanting," low "liking."

Ref: Berridge, K.C. & Robinson, T.E. (2003). "Parsing reward." *Trends in Neurosciences*, 26(9), 507–513.

6.1.1 Incentive Salience (Wanting)

$$W(s,t) = V(s) \cdot \kappa_{\text{DA}}(t) \tag{45}$$

where $V(s)$ is learned value and $\kappa_{\text{DA}}(t)$ is current DA state (amplifier):

$$\kappa_{\text{DA}}(t) = 1 + k_{\text{sens}} \cdot ([\text{DA}]_t - [\text{DA}]_{\text{baseline}}) \tag{46}$$

6.1.2 Hedonic Pleasure (Liking)

$$L = L_{\text{opioid}} + L_{\text{cannabinoid}} \quad \text{(independent of DA)} \tag{47}$$

6.1.3 Addiction: Sensitization of Wanting

$$k_{\text{sens}} \uparrow \text{ with repeated drug use} \tag{48}$$

$$W^{\text{addicted}} = V(s) \cdot \kappa_{\text{DA}}^{\text{sensitized}} \gg W^{\text{naive}} \tag{49}$$

This explains drug craving: sensitized κ_{DA} amplifies incentive salience for drug cues even when L (liking) is diminished by tolerance.

6.2 Addiction Allostatic Shift

What it models: Chronic drug use depletes hedonic set point through anti-reward systems (CRH, dynorphin, NE). The new allostatic baseline is below normal → dysphoria during abstinence → compulsive drug use to relieve negative state.

Ref: Koob, G.F. & Le Moal, M. (2008). "Addiction and the brain antireward system." *Annual Review of Psychology*, 59, 29–53.

6.2.1 Allostatic Hedonic Set Point

$$H^*(t) = H_0 - \int_0^t k_{\text{anti-reward}} \cdot f_{\text{drug}}(\tau)\, d\tau \tag{50}$$

6.2.2 Withdrawal Negative State

$$\text{Dysphoria}(t) = H^*(t) - H_0 < 0 \tag{51}$$

Variable	Definition	Values
H_0	Normal hedonic baseline	0 (normalized)
$H^*(t)$	Shifted set point (increasingly negative)	—
$k_{\text{anti-reward}}$	Rate of anti-reward recruitment	—
$f_{\text{drug}}(\tau)$	Drug intake history	—

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7 Fear and Threat Circuits

7.1 Fear Conditioning — LTP at BLA Synapses

What it models: CS (tone) → BLA synaptic strengthening via NMDA-dependent LTP when paired with US (shock). After conditioning, CS alone activates BLA→CeA→defensive outputs.

Ref: Rescorla, R.A. & Wagner, A.R. (1972). *Classical Conditioning II*, pp. 64–99.

7.1.1 Associative Strength Update

$$V_{\text{CS}}^{(n+1)} = V_{\text{CS}}^{(n)} + \alpha\beta \left(\lambda - \sum_k V_k^{(n)} \right) \tag{52}$$

7.1.2 BLA Synaptic Weight (LTP Mechanism)

$$\frac{dw_{\text{CS}}}{dt} = \eta_{\text{LTP}} \cdot [\text{Ca}^{2+}]_{\text{BLA}} \cdot x_{\text{CS}} - \eta_{\text{LTD}} \cdot w_{\text{CS}} \tag{53}$$

where Ca^{2+} influx through NMDA during US (US depolarizes → unblocks Mg^{2+} → Ca^{2+} influx).

7.2 Fear Extinction — vmPFC-ITC Inhibition of CeA

What it models: Extinction is new inhibitory learning (NOT erasure). Repeated CS-alone presentations activate vmPFC → intercalated cell masses (ITCs, GABAergic) → inhibit CeA output → suppress fear expression. This explains why extinction is context-dependent and fear can return.

Ref: Milad, M.R. & Quirk, G.J. (2002). "Neurons in medial prefrontal cortex signal memory for fear extinction." *Nature*, 420(6911), 70–74.

7.2.1 Extinction Learning Update

$$V_{\text{CS}}^{(n+1)} = V_{\text{CS}}^{(n)} + \alpha\beta(0 - V_{\text{CS}}^{(n)}) = V_{\text{CS}}^{(n)}(1 - \alpha\beta) \tag{54}$$

(US absent → $\lambda = 0$ → prediction error $= -V_{\text{CS}}$)

7.2.2 vmPFC→ITC→CeA Inhibition

$$r_{\text{CeA}}(t) = r_{\text{BLA}}(t) \cdot (1 - w_{\text{ITC}} \cdot r_{\text{vmPFC}}(t)) \tag{55}$$

7.2.3 Fear Renewal (Context-Dependence)

$$r_{\text{CeA}}^{\text{renewal}} = r_{\text{BLA}} - w_{\text{ITC}} \cdot r_{\text{vmPFC}} \cdot f_{\text{context}} \tag{56}$$

where $f_{\text{context}} = 1$ in extinction context, $f_{\text{context}} \approx 0$ in original (fear) context.

Variable	Definition
V_{CS}	CS associative strength (fear)
w_{ITC}	vmPFC-driven ITC inhibition strength
r_{vmPFC}	vmPFC (infralimbic) firing rate
f_{context}	Context gate (1 = extinction context)

Key insight: Fear returns after extinction because the original BLA→CeA trace is not erased — it is suppressed by vmPFC-ITC inhibition. In PTSD, vmPFC (infralimbic cortex) is hypoactive → ITC suppression is impaired → CeA remains hyperactive even after extinction training.

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8 Hippocampal Memory Circuits

8.1 Pattern Separation — Sparse Coding in Dentate Gyrus

What it models: The dentate gyrus (DG) converts overlapping cortical input patterns into sparse, orthogonal representations, minimizing interference between similar memories.

Ref: O'Reilly, R.C. & McClelland, J.L. (1994). "Hippocampal conjunctive encoding, storage, and recall: avoiding a trade-off." *Hippocampus*, 4(6), 661–682.

8.1.1 Sparseness Measure

$$a = \frac{N_{\text{active}}}{N_{\text{total}}} \tag{57}$$

DG: $a \approx 0.02$–0.05 (very sparse); Entorhinal cortex: $a \approx 0.15$–0.25 (dense)

8.1.2 Pattern Overlap (Similarity Before and After DG)

$$\rho_{\text{input}} = \frac{\mathbf{p}_1 \cdot \mathbf{p}_2}{|\mathbf{p}_1||\mathbf{p}_2|} \tag{58}$$

$$\rho_{\text{DG}} = \frac{\mathbf{q}_1 \cdot \mathbf{q}_2}{|\mathbf{q}_1||\mathbf{q}_2|} \tag{59}$$

Pattern separation: $\rho_{\text{DG}} \ll \rho_{\text{input}}$ (orthogonalization)

8.1.3 Winner-Take-All Mechanism

$$r_i^{\text{DG}} = \begin{cases} r_i & \text{if } r_i \in \text{top } k\% \\ 0 & \text{otherwise} \end{cases} \tag{60}$$

8.2 Pattern Completion — CA3 Attractor Network

What it models: CA3 recurrent collaterals allow retrieval of complete memories from partial cues (Hopfield attractor network). A degraded input cue converges to the nearest stored pattern.

$$s_i(t+1) = \text{sign}\left(\sum_{j \in \text{stored}} w_{ij} s_j(t) + \text{noisy cue}_i \right) \tag{61}$$

8.2.1 Retrieval Error Rate

$$P_{\text{error}} \approx \exp\left(-\frac{(1-f)^2}{2f(1-f)} \cdot N \right) \tag{62}$$

where f = pattern activity fraction, N = CA3 neurons ($\sim$ 300,000).

8.3 Memory Consolidation — Sharp-Wave Ripple Replay

What it models: During NREM sleep, hippocampal sharp-wave ripples (100–250 Hz) replay recent experience in compressed time ($\sim 20\times$ faster). This drives cortical slow oscillations to consolidate memories.

Ref: Nádasdy, Z., Hirase, H., Czurkó, A., et al. (1999). "Replay and time compression of recurring spike sequences in the hippocampus." *Journal of Neuroscience*, 19(21), 9497–9507.

8.3.1 Temporal Compression Ratio

$$\tau_{\text{replay}} = \frac{\tau_{\text{encoding}}}{\xi} \tag{63}$$

where $\xi \approx 20$ (compression factor; awake theta: 8 Hz replay $\rightarrow$ SWR: ~ 160 Hz equivalent)

8.3.2 Replay Fidelity (Pearson Correlation)

$$r_{\text{replay}} = \frac{\text{Cov}(\text{awake order}, \text{replay order})}{\sigma_{\text{awake}}\sigma_{\text{replay}}} \tag{64}$$

Empirically: $r_{\text{replay}} \approx 0.5$–$0.9$ (forward) and ≈ -0.3–-0.6 (reverse replay)

8.3.3 Coupled Oscillation (3-Way Coordination)

$$\phi_{\text{spindle}}(t) \subset \phi_{\text{slow-osc up-state}}(t) \subset \phi_{\text{SWR}}(t) \tag{65}$$

Sleep spindles (12–16 Hz) are nested within cortical slow oscillation up-states (~ 0.75 Hz), which in turn are coupled with hippocampal SWRs. This triple nesting provides the precise timing for systems consolidation.

Variable	Definition	Values
ξ	Temporal compression ratio	$\sim 20\times$
r_{replay}	Replay correlation (forward/reverse)	0.5–0.9
SWR frequency	Ripple band	100–250 Hz
Spindle frequency	Sleep spindles	12–16 Hz
Slow oscillation	Cortical SO	0.5–1 Hz

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9 Case Studies

9.1 Case Study 1: Hopfield Network — Memory Capacity Test

Scenario: A Hopfield network has $N = 100$ neurons. How many patterns can be stored reliably? If we try to store $p = 20$ patterns, what is the retrieval success probability?
Maximum capacity:

$$p_{\max} = 0.138 \times N = 0.138 \times 100 = 13.8 \approx 13 \text{ patterns} \tag{66}$$

For $p = 20$ patterns (above capacity):

Using Hopfield's (1982) noise analysis:

$$\text{SNR} = \frac{N}{p} = \frac{100}{20} = 5 \tag{67}$$

Expected retrieval error with $p = 20$ above $p_{\max}$: retrieval fails to converge to stored patterns $\sim$ 15–20% of time.

Result:

- Safe storage limit: **13 patterns** (reliable retrieval from partial cues)
- At 20 patterns: $\sim$ 80% correct retrieval (some spurious attractors appear)
- At 30 patterns: catastrophic forgetting ($\sim$ 50% errors)

Biological implication: DG pattern separation reduces effective overlap before CA3 storage, allowing CA3 to store more functionally independent patterns.

9.2 Case Study 2: E-I Balance — Gamma Oscillation Frequency

Scenario: A cortical column has $\tau_E = 5$ ms, $\tau_I = 10$ ms. Connectivity constants: $c_1 = 12$ (E→E), $c_2 = 8$ (I→E), $c_3 = 12$ (E→I), $c_4 = 3$ (I→I). Sigmoid slope $a = 0.4$ at operating point. What is the oscillation frequency?

Jacobian elements:

$$J_{EE} = c_1 S'_E/\tau_E = 12 \times 0.4/5 = 0.96 \text{ ms}^{-1} \tag{68}$$

$$J_{II} = -c_4 S'_I/\tau_I = -3 \times 0.4/10 = -0.12 \text{ ms}^{-1} \tag{69}$$

$$J_{EI} = -c_2 S'_I/\tau_E = -8 \times 0.4/5 = -0.64 \text{ ms}^{-1} \tag{70}$$

$$J_{IE} = c_3 S'_E/\tau_I = 12 \times 0.4/10 = 0.48 \text{ ms}^{-1} \tag{71}$$

Eigenvalues:

$$\lambda_{1,2} = \frac{(0.96 - 0.12)}{2} \pm \frac{1}{2}\sqrt{(0.96 + 0.12)^2 + 4(-0.64)(0.48)} \tag{72}$$

$$= 0.42 \pm \frac{1}{2}\sqrt{1.166 - 1.229} = 0.42 \pm \frac{1}{2}\sqrt{-0.063} \tag{73}$$

$$= 0.42 \pm 0.126i \text{ ms}^{-1} \tag{74}$$

Oscillation frequency:

$$f = \frac{\text{Im}(\lambda)}{2\pi} = \frac{0.126 \text{ ms}^{-1}}{2\pi} \approx \frac{126 \text{ s}^{-1}}{2\pi} \approx 20 \text{ Hz} \quad \text{(beta band)} \tag{75}$$

Result: This network oscillates at $\sim$ 20 Hz (beta rhythm). To obtain gamma ($\sim$ 40 Hz), reduce τ_I to $\sim$ 5 ms (faster PV+ interneurons) or increase c_3 (stronger E→I drive).

9.3 Case Study 3: Fear Conditioning and Extinction Trials

Scenario: A rat receives 10 fear conditioning trials (CS-US), then 10 extinction trials (CS only). Track V_{CS} throughout.
Parameters: $\alpha = 0.4$, $\beta = 0.3$, $\lambda_{\text{conditioning}} = 1.0$, $\lambda_{\text{extinction}} = 0$
Conditioning Phase (trials 1–10):

$$\Delta V = 0.12 \times (1.0 - V) \tag{76}$$

Trial	V	δ
1	0.000	1.000
3	0.217	0.783
5	0.397	0.603
7	0.545	0.455
10	0.718	0.282

Extinction Phase (trials 11–20):

$$\Delta V = 0.12 \times (0 - V) = -0.12V \tag{77}$$

Trial	V	δ
11	0.718	−0.718
13	0.558	−0.558
15	0.434	−0.434
18	0.297	−0.297
20	0.231	−0.231

Result: After 10 conditioning + 10 extinction: $V = 0.231$ (suppressed but not zero).
Biological note: Extinction never fully erases the original memory (V never reaches 0 in this model because the decrement scales with remaining V). This is consistent with clinical observation: fear can return after extinction (spontaneous recovery, reinstatement, renewal). The persistent $V > 0$ represents the original BLA trace that vmPFC-ITC actively suppresses rather than erases.

9.4 Case Study 4: Small-World Brain Network

Scenario: A brain network has $N = 90$ regions (AAL atlas). Random rewiring gives $C_{\text{random}} = 0.018$ and $L_{\text{random}} = 2.8$. The observed brain network has $C_{\text{obs}} = 0.44$ and $L_{\text{obs}} = 3.1$. Is it small-world?
Small-world index:

$$\sigma = \frac{C_{\text{obs}}/C_{\text{random}}}{L_{\text{obs}}/L_{\text{random}}} = \frac{0.44/0.018}{3.1/2.8} = \frac{24.4}{1.11} = 22.0 \tag{78}$$

Result: $\sigma = 22 \gg 1 \Rightarrow$ **Strongly small-world.**
Interpretation:

- $C_{\text{obs}}/C_{\text{random}} = 24.4$: The brain is 24× more locally clustered than random (specialized modules/networks)

- $L_{\text{obs}}/L_{\text{random}} = 1.11$: Path length is only 11% longer than random (efficient global communication)
- Any brain region can reach any other in ~ 3 synaptic steps on average

This explains how the brain achieves both specialized processing (high clustering = visual cortex, motor cortex, etc.) and integrated consciousness (short paths = rapid cross-network communication).

9.5 Case Study 5: Cerebellar Learning — Purkinje Cell Weight Convergence

Scenario: A Purkinje cell receives 5 parallel fiber inputs with activities $\mathbf{x} = [0.8, 0.3, 0.9, 0.1, 0.6]$. The desired output is $d = 0$, current weights $\mathbf{w} = [0.5, 0.5, 0.5, 0.5, 0.5]$, $\eta = 0.1$.
Step 1 — Current output:

$$y = \sum_i w_i x_i = 0.5(0.8 + 0.3 + 0.9 + 0.1 + 0.6) = 0.5 \times 2.7 = 1.35 \tag{79}$$

Step 2 — Error:

$$e = d - y = 0 - 1.35 = -1.35 \tag{80}$$

Step 3 — Weight update ($\Delta w_i = -\eta x_i e$):
Wait — the error is negative (output too high, need to decrease), so weights should DECREASE:

$$\Delta w_i = -\eta x_i e = -0.1 \times x_i \times (-1.35) = +0.135 x_i \tag{81}$$

This increases weights — contradiction. The Marr-Albus rule causes LTD (weakening):

$$\Delta w_i = -\eta x_i |e| = -0.1 \times x_i \times 1.35 \tag{82}$$

Correct LTD updates:

i	x_i	Old w_i	Δw_i	New w_i
1	0.8	0.50	-0.108	0.392
2	0.3	0.50	-0.041	0.459
3	0.9	0.50	-0.122	0.378
4	0.1	0.50	-0.014	0.486
5	0.6	0.50	-0.081	0.419

New output after 1 update:

$$y_{\text{new}} = 0.392(0.8) + 0.459(0.3) + 0.378(0.9) + 0.486(0.1) + 0.419(0.6) = 0.801 \tag{83}$$

Error reduced: $|e| : 1.35 \to 0.80$ (41% reduction in one trial).

9.6 Case Study 6: SWR Memory Replay — Compression Calculation

Scenario: A rat runs a 2-meter linear track in 4 seconds (encoding). During subsequent SWR, the same place cell sequence replays in 100 ms. Calculate compression ratio and verify consistency with theta/SWR frequencies.
Temporal compression:

$$\xi = \frac{\tau_{\text{encoding}}}{\tau_{\text{replay}}} = \frac{4000 \text{ ms}}{100 \text{ ms}} = 40\times \tag{84}$$

Place cell firing during encoding:

- Track = 2 m; $\sigma_{\text{place}} = 20$ cm $\rightarrow$ 10 distinct place fields activated sequentially
- Each active for $\sim$ 400 ms (4 s / 10 fields)
- Theta frequency during running = 8 Hz $\rightarrow$ 3.2 theta cycles per field

During SWR replay:

- Each place cell active for $\sim$ 10 ms (100 ms / 10 fields)
- SWR frequency = $\sim$ 200 Hz $\rightarrow$ place cells fire at gamma-locked ($\sim$ 40 Hz) bursts within SWR

Comparison with theta:

- Theta encoding: 8 Hz $\rightarrow$ 125 ms/cycle
- SWR replay: $\sim$ 200 Hz $\rightarrow$ 5 ms/cycle
- Compression = 125/5 = **25**$\times$ (within SWR burst; total 40$\times$ including between-cell intervals)

Result: Replay compresses 4-second experience into 100 ms — a **40**$\times$ **compression**. This allows $\sim$ 100–200 events to be replayed per night during NREM sleep, supporting memory consolidation for the day's experiences.

Next: Chapter 6 — Sensory Processing